

View Reviews

Paper ID

3864

Paper Title

A Cascaded DERNET and YOLO11 Framework for Spinal Lesion Triage and Localization with Explainable AI

Reviewer #1

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines: <https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

MIC

4. How would you describe the paper?

Methodological contribution

5. Please describe the main contribution of the paper.

The authors propose a framework for Spinal lesion detection and location in 2D radiographic images. Authors propose a two-step method, with a first step to do binary triage (healthy / pathological) and a second step in which the lesion is located in the image.

An ensambling of 3 networks is proposed for the binary decision, then a Yolo variant is used to locate the lesions. Existing explainability methods are used to provide feedback to the clinician.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

- the paper is clearly written

- the obtained results outperform the state of the art in the VinDr-SpineXR dataset

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

- the major weakness is that several claims are not backed up with an experiment or an ablation study. For example:
- 3.2.(b) "This mechanism specifically inhibits non-diagnostic background noise and anatomical overlap (...) eliminate false positives with mid-scale lesions" - mid-scale lesions were not defined, and no in depth quantitative analysis is provided to show the claim: an ablation with/without the EfficientNetV2.
- 3.3 (b) "... block ensures that fine-grained pathology signals are preserved across deep layers, neutralizing the severe class imbalance inherent to spinal benchmarks". The class imbalance is not quantitatively described, it is unclear to what authors refer to with "fine-grained pathology signals" without showing an image or example.
- 4.2 "At the lesion level, imbalance was further mitigated through focal loss optimization and targeted copy-paste augmentation..." but no ablation with / without augmentation is provided.
- the technical contribution of the proposed approach is relatively weak: an weighted ensembling method of existing architectures for the binary classification, a retrained Yolo for the detection, and the use of existing explainability methods (LIME, Grad-Cam, and Qualitative visualization) to have an interpretable layer.
- It is not clear how parameter optimization was performed: Page 5 first paragraph: "We dynamically establish the optimal classification threshold ... by maximizing the macroscopic F1-score .." on which dataset was this performed? On the final test-set? VinDR only provides a train / test set. Did authors create a validation set for parameter tuning?
- authors will release a list of "additional resources", such as "real-time demos" and "reproducibility resources", but "source code" is not mentioned.

8. Please rate the clarity and organization of the paper.

Satisfactory

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

The authors claimed to release the source code and/or dataset upon acceptance of the submission.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

No

12. Optional: If you have any additional comments to share with the authors, please provide them here. Please also refer to our Reviewer's guide on what makes a good review and pay specific attention to the different assessment criteria for the

different paper categories: <https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

- In Tab 2: LT6 and LT11 values are highlighted in bold for the method, but they are not the best in the column... for LT13, 5.30 should be the bold value.

- The CSP DarkNet ([A] and/or [B]) citation should be added

[A] Wang, Chien-Yao, Hong-Yuan Mark Liao, Yueh-Hua Wu, Ping-Yang Chen, Jun-Wei Hsieh, and I-Hau Yeh. "CSPNet: A new backbone that can enhance learning capability of CNN." In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops, pp. 390-391. 2020.

[B] Bochkovskiy, Alexey, Chien-Yao Wang, and Hong-Yuan Mark Liao. "Yolov4: Optimal speed and accuracy of object detection." arXiv preprint arXiv:2004.10934 (2020).

- Fig 4 captions are way too small to be read.

- The title DerNet is already used [C]. Maybe consider changing the name.

[C] Peng, Zhengyao, Lin Yin, Zewen Sun, Qian Liang, Xiaopeng Ma, Yu An, Jie Tian, and Yang Du. "DERnet: a deep neural network for end-to-end reconstruction in magnetic particle imaging." Physics in Medicine & Biology 69, no. 1 (2024): 015002.

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

3. Weak Reject — marginally below the acceptance threshold, but would not mind if accepted, dependent on rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

I'm hesitant about this paper:

On the one hand, it outperforms the state of the art, and the authors promise to share the project resources (note that code is not explicitly mentioned).

On the other hand, no technical contribution is provided, and it is unclear whether the parameter tuning was performed to maximize test-set values rather than an independent validation set.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

Confident but not absolutely certain (3)

Reviewer #2

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines: <https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

MIC

Clinical Translation of Methodology

4. How would you describe the paper?

Methodological contribution

5. Please describe the main contribution of the paper.

The paper proposes a cascaded deep learning framework for spinal lesion analysis. The key idea is to address limitations including: (a) severe class imbalance, (b) small region of interest (i.e. lesion size). The paper also incorporates explainability using established LIME, Grad-CAM techniques.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

The paper addresses a clinically relevant problem. Though limited, but papers reflect awareness of clinical interpretability requirements.

Proposed framework demonstrates effectiveness under extreme class imbalance.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

1. The ensemble weights $\Omega = [0.42, 0.32, 0.26]$ are mentioned but their deviation is not explained. How were they learned (grid search or manually)?

2. Understandable that the mAP@0.5 is reported for benchmarking but it's insufficient for medical object detection.

3. Single dataset used, no external validation (has been acknowledged as limitation).

4. Choices are arbitrary: why YOLO11-L over other variants? Did the authors test other CNN variants?

5. LIME and Grad-CAM are used only qualitatively.

Writing inconsistencies (for 3.2 (c), it should be $w_3 = 0.26$, instead of w_2 .)

Figure 4c, visuals are very low-resolution to critically evaluate.

8. Please rate the clarity and organization of the paper.

Satisfactory

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

The submission has provided an anonymized link to the source code, dataset, or any other dependencies.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

3. Weak Reject — marginally below the acceptance threshold, but would not mind if accepted, dependent on rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

The contribution appears more engineering integration than methodological innovation. No ablation showing sensitivity to weights. No statistical significance testing or confidence intervals.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

Very confident (4)

Reviewer #4

Questions

3. Please categorize the relevance of the paper (you may choose more than one). Note the different assessment criteria for the different paper categories as outlined in the Reviewer Guidelines: <https://conferences.miccai.org/2026/en/REVIEWER-GUIDELINES.html>

MIC

Clinical Translation of Methodology

4. How would you describe the paper?

Both

5. Please describe the main contribution of the paper.

This paper's main contribution is a cascaded and interpretable deep learning framework for spinal lesion screening and localization on radiographs. The method combines a high-sensitivity ensemble classifier (DERNet) for triage with a customized YOLO11-L detector for lesion localization, aiming to better handle severe class imbalance and small lesion size. In addition, the paper emphasizes interpretability by incorporating LIME, Grad-CAM, and qualitative visualizations, and demonstrates improved performance on the VinDr-SpineXR benchmark.

6. Please list the major strengths of the paper. For example, you could highlight a novel formulation, an original way to use data, a demonstration of clinical feasibility, a novel application, a particularly strong evaluation, or anything else that is a strong aspect of this work. Please provide details, for instance, if a method is novel, explain what aspect is novel and why this is interesting.

The major strengths of this paper are as follows:

1. The paper addresses a clinically important problem. Automated analysis of spinal radiographs is highly valuable for early screening and triage, especially because many spinal lesions are visually subtle, small in scale, and easily obscured by overlapping anatomical structures. The paper clearly explains why this is a challenging and clinically meaningful task, rather than merely a benchmark problem.
2. The localization stage is designed in a task-aware manner. The paper does not treat spinal lesion detection as a generic object detection problem, but explicitly takes into account that many lesions occupy less than 1% of the field of view. Accordingly, the localization stage is tailored for this setting. The customized YOLO11-L detector adopts a CSP-Darknet backbone, C2PSA attention, PANet-based multi-scale fusion, and multi-scale detection heads to better support precise localization of subtle lesions. The use of focal loss tuning and copy-paste augmentation also reflects a strong awareness of the severe foreground/background imbalance in this task.
3. The empirical performance is strong relative to the reported baselines. DERNet achieves an AUROC of 91.03% for image-level classification, outperforming the listed baselines, while the YOLO11-L localization stage achieves a mAP@0.5 of 40.10%, again surpassing the compared detectors on the VinDr-SpineXR benchmark.
4. The paper shows awareness of practical deployment considerations. The discussion of latency optimization, an ONNX-accelerated inference engine, and reproducibility-oriented resources suggests that the authors are not only concerned with offline benchmark performance, but are also thinking about the model as a potentially usable clinical decision-support tool. Although this aspect still requires stronger quantitative validation, the deployment-oriented perspective is itself a positive aspect of the work.

7. Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

The major weaknesses of this paper are as follows:

1. The methodological novelty is limited and mainly lies at the system-integration level. The overall framework is reasonable, but many of its core components are standard: the triage stage is a weighted ensemble of established backbones (DenseNet121, EfficientNetV2-S, and ResNet50); the localization stage is a customized YOLO-style detector; and the interpretability part relies on standard post-hoc tools such as LIME and Grad-CAM. In addition, prior work already exists on spinal lesion classification/detection and on the VinDr-SpineXR benchmark itself, including the VinDr baseline framework, generic two-stage detectors such as Faster R-CNN and Sparse R-CNN, and spinal lesion detectors such as EGCA-Net. Therefore, the main contribution of this paper is better characterized as an engineering integration of known components within a cascaded framework, rather than a substantially novel methodological advance.
2. The paper does not evaluate the cascaded system in a truly end-to-end manner. The manuscript reports the classification performance of DERNet and the detection performance of YOLO11-L separately, but it does not clearly quantify the performance of the full cascaded pipeline after the triage gate. Since the detector only processes cases that are passed by the classifier, the final lesion-level recall/sensitivity of the system depends on the combined effect of both stages. Without end-to-end evaluation, it is difficult to accurately assess the practical clinical utility of the proposed pipeline.
3. There is insufficient ablation analysis for the main design choices. Although the paper presents the cascaded design as a solution to class imbalance and the sensitivity-specificity trade-off, it does not provide sufficient ablations to validate these claims. For example, there is no direct comparison with a single-stage detector-only pipeline, no analysis of the contribution of the ensemble weights, and no threshold-sensitivity analysis for the DERNet gate. Likewise, the customized YOLO11-L modifications (such as C2PSA, PANet, and copy-paste augmentation) are not studied component by component. This makes it difficult to determine which parts of the performance gain come from the cascade itself and which come from standard optimization choices.
4. The interpretability analysis is mostly qualitative and does not strongly support the claim of explainability. The use of LIME and Grad-CAM is a positive addition, but these are standard post-hoc visualization tools rather than novel interpretability methods. More importantly, the paper only provides qualitative examples and does not include quantitative evaluation, expert validation, or systematic analysis to show that these explanations are reliable or clinically valuable. Therefore, the claim of “explainable AI” is only partially substantiated.
5. Despite the paper’s emphasis on class imbalance, performance on rare classes remains weak. The manuscript presents the framework as a way to address severe class imbalance, but Table 2 shows that performance on some rare or heterogeneous categories is still very limited; for example, LT13 (“other lesions”) achieves an AP of only 2.99%. This

suggests that the long-tail problem is far from solved, and that the paper's claims about substantially mitigating severe class imbalance may be somewhat overstated.

8. Please rate the clarity and organization of the paper.

Satisfactory

9. Please comment on the reproducibility of the paper. Please be aware that providing code and data is a plus, but not a requirement for acceptance.

The submission does not mention open access to source code or data, but provides a clear and detailed description of the algorithm to ensure reproducibility.

10. Code of Ethics Check. Based on your review and your understanding of the MICCAI Scientific Code of Ethics, do you believe this submission may involve a potential ethics concern or violation? Reviewers are expected to familiarize themselves with the MICCAI Scientific Code of Ethics before completing this assessment: <https://miccai.org/index.php/about-miccai/policies/scientific-code-of-ethics/>

No

13. Rate the paper on a scale of 1-6, 6 being the strongest (6-4: accept; 3-1: reject). Please use the entire range of the distribution. Spreading the score helps create a distribution for decision-making.

3. Weak Reject — marginally below the acceptance threshold, but would not mind if accepted, dependent on rebuttal

14. Please justify your recommendation. What were the major factors that led you to your overall score for this paper?

My overall score was based on a balance between the paper's practical strengths and its current limitations. On the positive side, the paper addresses a clinically meaningful problem, proposes a reasonable cascaded formulation for spinal radiograph triage and localization, and reports strong empirical results on the VinDr-SpineXR benchmark, including 91.03% AUROC for image-level classification and 40.10% mAP@0.5 for lesion-level detection. The task-aware design of the localization stage and the inclusion of interpretability tools are also positive aspects.

However, the main factors limiting my score are that the methodological novelty appears somewhat limited and is mainly at the system-integration level, the cascaded framework is not evaluated in a truly end-to-end manner, and the paper does not provide sufficient ablation studies to validate its key design choices. In addition, the explainability analysis is mostly qualitative and does not yet strongly support the "explainable AI" claim, while performance on rare classes remains weak despite the paper's emphasis on handling class imbalance.

16. In view of your answers above and your overall experience, how would you rate your confidence in your review?

Very confident (4)